Big Data processing and Analytics with MongoDB, PySpark and Tableau/Power BI.

The project should involve data processing, querying, analysis and visualization using PySpark, MongoDB and Tableau/Power BI.

Group project:

- The project shall be carried out as a group of two-three members.
- Should be working with using big data tools and technologies for collecting, managing, analyzing and visualizing big data in any domain.
- Performing whole big data management life cycle.

Technology usage:

- PySpark, MongoDB and Tableau/ Power BI must be used. Coding should be done in python.

Grading:

- The project will be graded on:
  - performing whole big data management life cycle
  - the quality of the produced code
  - effort
  - originality
  - project report (see report outline below), and
  - the development processes.

Code:

- You should comment the code lightly in line. This does not mean one comment line for each code line but, e.g., one brief comment for each logical group of code lines.

Project Report:

- The project should be written in the project template given to you and can be downloaded from Canvas.
- The project report file should be anonymous, and contain the exam number of your group, BOTH on the first page and in the file name (e.g., PGR108ProjectReport25_108_113.pdf .)
- It must be in English (STRICT REQUIREMENT).

In the report explain:

  - Introduction:
    - You should introduce the domain and explain the purpose of your project. (motivation)

        - Why have you worked on this project? (problem)
        - Project objective/goal
        - Why is it a good idea to do this using other technologies?
        - What can you do now that wasn't possible before?
        - Limitations and assumptions
        - Project management (You should describe the contributions each person in your group has made to the project)
    - Background and literature review:
        - You should list the big data technologies/tools/standards you have used and explain briefly why you chose each of them.
        - Did you consider alternatives?
        - Why were the ones you chose better?
    - Existing literature on the project (if any).
    - Research Methodology:
        - Describe your dataset.
        - Data preprocessing
        - Big data life cycle
            - If you are reading/converting/lifting data from multiple sources and/or using existing tools in addition to your own program, you should probably include a flow chart or architecture sketch (which is different from a class diagram).
    - Results and discussion
        - Describe/report your results and discuss them.
    - Conclusions
        - You should mention any particular problems you have had and/or things you want to do differently next time.
        - Do not include code snippets in the report (not tutorial!)

    - If you want to briefly describe how to run the code you have submitted, you can do that separately in a README.TXT file.

Project list:

1. JUCars-2024: A dataset of used car listings in Jordan - ScienceDirect
2. BigFlow-NIDS: A large-scale dataset for network intrusion detection in big data environment - ScienceDirect
3. Structured over-the-counter medicine dataset to improve the safety of pharmaceuticals - ScienceDirect
4. Air quality monitoring at Universidad Iberoamericana, Mexico City: Ozone and meteorological data during the rainy and ozone seasons (Mar–Nov 2024) - ScienceDirect

5. A fully synthetic textual dataset of student learning habits and preferences generated using a large language model - ScienceDirect
6. A multi-institutional dataset on patient-individual lymph node involvement in hypopharyngeal and laryngeal squamous cell carcinoma - ScienceDirect
7. Order picking dataset from a warehouse of a footwear manufacturing company - ScienceDirect
8. A dataset on the segregation of students with disabilities in Brazil - ScienceDirect
9. Weather and electrical demand and consumption data of a small Mexican community - ScienceDirect
10. Indonesian pharmaceutical dataset for self-medication - ScienceDirect
11. Thalassemia dataset covering clinical, socioeconomic, and mental health aspects - ScienceDirect
12. A dataset on vulnerabilities affecting dependencies in software package managers - ScienceDirect
13. Dataset on motivations and perceptions regarding teaching as a career among teacher education students - ScienceDirect
14. Dataset from the Indonesian adaptation of the center for epidemiologic studies depression scale for emerging adults - ScienceDirect
15. A dataset on the use of online video by students at the in-video level - ScienceDirect
16. AI adoption in Spain (2023–2025): A web-derived dataset based on LLMs - ScienceDirect
17. Dataset on electricity usage measurement for lower-to-middle-income primary and secondary schools in Western Cape, South Africa - ScienceDirect
18. Database of 2025 Night of the Museums events in Poland - ScienceDirect
19. Building material price data for predictive cost estimation in construction - ScienceDirect
20. A dataset on adolescent mental health in Kenya - ScienceDirect
21. Dataset on passenger experience after riding an autonomous ferry - ScienceDirect
22. Cross-national survey data on student attitudes toward artificial intelligence - ScienceDirect
23. Nordic textile anatomy database: Composition of garments available in the nordic retail mass market and post-consumer textile waste market - ScienceDirect